

3D clustering on exabyte datasets for particle detectors: Eliminating background at the source

Project goals We propose a new reconstruction strategy for particle collider detectors, using machine-learning (ML)-based image-processing methods to perform **local three-dimensional pattern recognition**, replacing the two-dimensional clustering step in conventional workflows. This allows rejection of background signals at the earliest possible stage, before the computationally expensive track-finding and fitting algorithms are invoked. The combinatorial scaling of current algorithms suggests a potential speedup of an order of magnitude or more for current detectors. This approach will yield immediate scientific impact by reducing latency in reconstruction workflows for exabyte-scale datasets from years to months, while also improving reconstruction precision and discovery reach. It will lay the foundation for future projects in the context of the DOE Genesis mission, where MIT is involved in applying Foundation Models to detector-data reconstruction and in the development of 3D-stacked silicon tracking detectors.

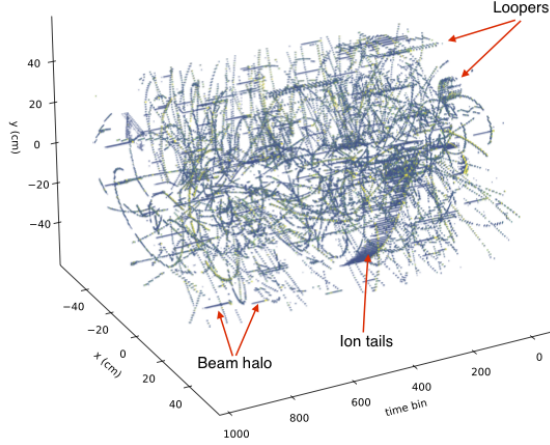


FIG. 1. sPHENIX TPC event display for proton-proton collisions at the Relativistic Heavy Ion Collider, showing trajectories of charged particles traversing the TPC gas volume. The TPC is situated inside a 1.5 T solenoidal magnetic field.

Particle tracking detectors Tracking detectors that reconstruct three-dimensional trajectories of charged subatomic particles have been at the heart of many of the most important discoveries over the last century. These devices led to a dozen Nobel Prize-winning advances, from the first observation of antiparticles to the Higgs boson discovery by the ATLAS and CMS experiments in 2012. Time Projection Chambers are particularly prominent

in current particle and nuclear physics experiments. Introduced by Nygren as a truly 3D tracking detector [1], TPCs are especially powerful for the study of the complex final states produced in high-energy nuclear collisions, where we investigate novel states of matter under conditions of extreme temperature, $T > 10^{12}$ K. An example of the 3D information provided by TPCs is shown in Fig. 1 for the sPHENIX TPC [2]. This device is an example of a streaming-readout TPC, reaching continuous multi-terabit-per-second data rates and producing multi-hundred-petabyte recorded data sets. These experiments therefore lie at the frontier of detector instrumentation and exascale scientific data processing. Large computer farms with $O(10^5-10^6)$ CPU cores are used to reconstruct these data. Even on these facilities, final processing times for the largest datasets reach years, creating a bottleneck for scientific discovery.

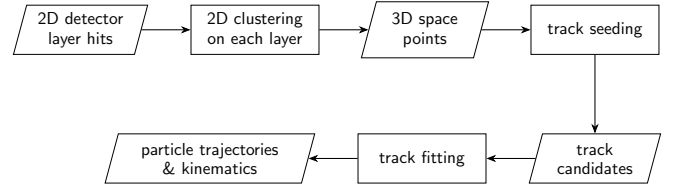


FIG. 2. Conventional charged-particle reconstruction workflow in a tracking detector. Computational cost is highly dependent on lower-level object multiplicity.

Particle collider detectors are often designed as a sequence of concentric cylindrical layers, with raw detector charge signals, or *hits*, located on these two-dimensional surfaces. Conventional reconstruction workflows, as in Fig. 2, employ several conceptually distinct stages that start from this 2D information. In a *clustering* step, hits from neighboring pixels on individual 2D layers are combined into *clusters*, marking the points where charged particles crossed the layer. The corresponding *space points* across layers are then connected during *track seeding* to form *track candidate trajectories*. Finally, Kalman-filter track fitting selects trajectories that are geometrically and kinematically consistent with physics-relevant charged particles.

Main challenge This layered workflow is successful and efficient in sparse environments. However, its total computational cost is strongly dependent on cluster multiplicity N , typically scaling as $O(N \log N)$. Close evaluation of sPHENIX TPC data reveals that large local cluster densities resulting from accelerator backgrounds, physics backgrounds, and detector artifacts lead to a combinatorial explosion in track seeding, which is tamed only in the highly selective, but computationally costly, track-fitting

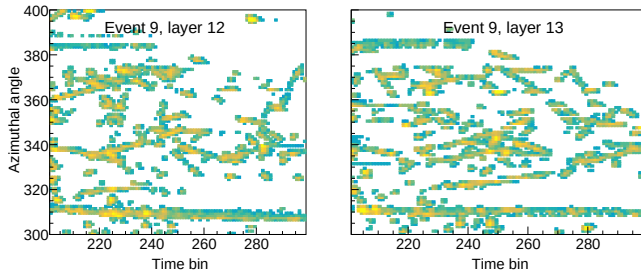


FIG. 3. Local TPC charge distributions in two consecutive layers for a high-occupancy Au+Au collision event. The persistence of correlated structures across layers illustrates the opportunity for three-dimensional pattern recognition.

stage. In essence, a large fraction of resources is spent processing clusters that contain no physics-relevant information. Several distinct background sources are readily visible in Fig. 1: low-momentum background electron “loopers,” “beam halo” particles unrelated to the collision, and large signal ion tails. The limitations of 2D clustering are evident in Fig. 3, which shows the 2D signal on corresponding patches of two consecutive detector layers. As the signals from particles of interest vary in amplitude, nearest-neighbor distance, and incident angle, an effective rejection of backgrounds at the 2D clustering stage is not feasible, leading to a large number of background space points being passed on for further processing. Only at the computationally most expensive track-fitting stage are these signals removed.

Proposed approach We argue that these same backgrounds are more readily recognized in raw detector coordinates, provided one moves from a sequential 2D treatment of input signals to an approach that processes multiple adjacent detector layers at once. Rather than forcing the data into a layered point representation and solving a difficult global ambiguity problem afterward, we treat the TPC response as a 3D pattern-recognition problem from the outset. For each TPC readout module, data can be represented as a local ϕ -bin by time-bin by layer volume. Connected physics-relevant charge structures can be identified with image-processing algorithms and characterized by learned or engineered features, removing irrelevant structures before the expensive stages of track building begin.

We foresee a staged approach for this project. First, we will develop a local 3D clustering framework for raw TPC data. Starting from module-level detector volumes, we will identify contiguous 3D charge “blobs” across multiple

detector layers. These will be classified as physics-related or characteristic backgrounds through ML algorithms trained on detector simulations and on the results of the current chain. In the initial implementation, this classification could be used to accept or reject space points from conventional clustering, reducing the combinatorial load on subsequent steps. Already this first step could be integrated into the sPHENIX production workflow, executed at scale to make more efficient use of computing resources, potentially saving billions of CPU-core hours.

In a second implementation step, full 3D clustering based on ML image-processing algorithms will be integrated into the track-seeding step. Rather than starting from large clouds of loosely constrained space points, track seeding will be built on calibrated compact local track stubs stretching across multiple detector layers, including full covariance matrices, so that later reconstruction stages can stitch them together directly with Kalman-style propagation. This will further reduce combinatoric overhead and improve kinematic resolution by de-weighting spurious or malformed 2D charge information. For tracking detectors, increased resolution directly translates into improved discovery potential.

Future prospects This work also connects naturally to emerging AI and hardware developments at MIT in the context of the DOE Genesis mission. MIT will join collaborators in the development of the FM4NPP foundation model [3] for reconstruction of particle detector data. This approach is currently trained on space points. Moving to 3D track stubs promises improved performance and easier training, and may widen the applicability of the model across different experimental facilities. The 3D tracking approach is also directly connected to a Genesis proposal for the development of self-contained, 3D-stacked, AI-enabled silicon tracking devices, proposed by G.M. Innocenti and MIT Lincoln Laboratory.

Summary We ask for support for the development and deployment of a conceptually new approach to particle-tracking detector reconstruction, using ML techniques to move the initial reconstruction step from two to three dimensions. This promises to shorten the reconstruction timescale for underlying exabyte-scale data sets from years to months while improving reconstruction precision. Beyond the immediate benefits for the sPHENIX discovery science program, a successful demonstration of local 3D charge-pattern recognition would create a bridge between algorithm development, AI-based large-scale scientific computing, and future on-detector intelligence. The combination of immediate scientific value and a foundation for future high-risk, high-reward efforts aligns closely with the goals of the Research Support Committee program.

[1] D. R. Nygren, *The time-projection chamber: A new 4 pi detector for charged particles*, Tech. Rep. SLAC-PEP-NOTE-144 (Stanford Linear Accelerator Center and Lawrence

Berkeley Laboratory, 1974).

[2] sPHENIX Collaboration, *sPHENIX Technical Design Report*, Tech. Rep. (2019).

- [3] D. Park *et al.*, (2025), [arXiv:2508.14087 \[cs.LG\]](#).